

Semiconductor Optical Single-Photon Detectors

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Abstract—Semiconductor optical single-photon detectors are solid-state photodiodes enabling the detection of individual visible photons. These are essential tools connecting the photonic and electronic domains at the quantum level. Among these semiconductor-based-detectors, single-photon avalanche diodes (SPADs) are widely implemented in the mainstream, due to their convenience at practical thermoelectrically cooled temperature, low-operating costs, and considerable high-performing detectors. Although there are challenges with the device dark count rates and after-pulsing issue, their fairly high detection efficiency (~50%) in the visible range for advancing quantum communication, cryptography, and low-light imaging.

Index Terms—avalanche diodes, single-photon detectors, photon detection efficiency, semiconductors.

I. INTRODUCTION

SINGLE-PHOTON AVALANCHE DIODES (SPADs) are photodiodes operated in a Geiger mode, so they act as single-photon detectors. A single incident photon with some probability triggers an electron avalanche that produces a “current pulse”, indicating a detection event. SPADs are practical thermoelectrically for light waves at the 950 nm to 1650 nm region in broadband communications and data transmission over optical fibers [7], over the years, various semiconductor materials have been implemented as absorbers for SPADs for different wavelength detection range.

The highest efficiency commercial devices are based on a thick, 180 μm diameter high-purity silicon absorber combined with an active quenching circuit and cooling in a single practical module. These devices offer single-photon sensitivity in the 400-1000 nm wavelength range and achieve a peak efficiency of 65% at approximately 650 nm [1].

II. OPERATIONAL PRINCIPLES OF SPADs

SPADs operate similarly to a p-n junction diode with an applied reverse-biased above the breakdown voltage [5]. In which, the bias applied to the avalanche photodiode is sufficiently large enough to cause the **Markovian impact ionization** gain mechanism [3]. When a photon is absorbed with energy that is higher than the bandgap, it interacts with valence electrons and elevates the electrons into the valence band. This generates electron-hole pairs in the space charge (depletion) region, the excess carriers are then subjected to strong acceleration by the electric field; the electrons are swept into the n region and the holes into the p region. In the process, these energetic carriers collide with more bound

electrons of further atoms in the material, thereby ionizing them and generating additional electron-hole pairs in a chain reaction. This avalanche flow of free charge carriers produces a **measurable macroscopic current** in the milliamp range [8].

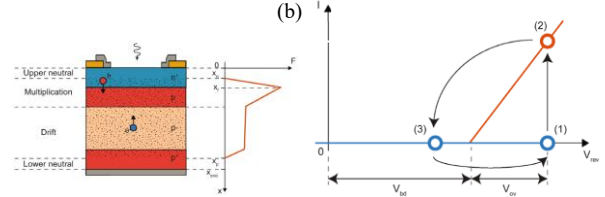


Fig. 1. a) Basic Structure of a SPAD, with corresponding electric field F . b) I-V characteristic of a SPAD; V_{rev} is the reverse bias; V_{bd} is the breakdown voltage, and V_{ov} is the overvoltage [2].

The typical I-V characteristics of a SPAD are sketched in Figure 1b [2]. As soon as a reverse bias $V_{\text{rev}} = V_{\text{bd}} + V_{\text{ov}}$ is applied (1), no free carriers are present in the depletion region and therefore no current is flowing. In this condition, the absorption of a single photon can initiate the impact ionization process giving rise to a self-sustaining avalanche current (2). A quenching circuit is used to reset the bias voltage (3).

In this scenario, single-photon sensitivity depends only on the detector since the macroscopic avalanche current that is generated by a SPAD can easily overcome the noise floor of the frontend electronics. As a result, the output of the readout electronics is typically a digital pulse corresponding to the detection of a photon. When only the information about a photon impinging on the detector is retrieved, the approach is usually referred to as photon counting. Nevertheless, SPADs can also provide information about the time of arrival of each photon with picosecond precision, a feature that is exploited in the so-called photon timing.

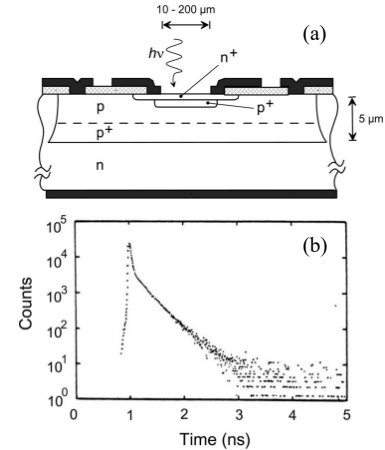


Fig. 2. (a) Cross-section of a double-epitaxial SPAD structure. (b) Photon arrival time distribution with this device [10].

III. CHARACTERISTICS OF SPADS

Photon detection efficiency (PDE) of a SPAD is a key parameter that describes the probability that a photon impinging on the active area of the detector succeeds in triggering an avalanche current that can be detected by electronics [2]. To have a macroscopic photogenerated current at the SPAD terminals, several conditions are needed: the photon must enter the device and be absorbed in its active layer; then, at least, one of the two released carriers (an electron-hole pair) must reach the high-field region and trigger the avalanche current [11-12]. The higher absorption coefficient, which depends on the wavelength and material, can increase the absorption probability of incident photons. And increasing the reverse voltage applied across SPADs' junctions can increase the triggering probability in the later condition, thus improving detection efficiency.

Dark count rate (DCR) is a critical noise parameter representing the average number of dark pulses per unit time. It is associated with the avalanche current in the SPAD is triggered, when the detector is kept in perfect dark conditions and without purposely sending any photons. The DCR can originate from various mechanisms, including band-to-band thermal generation, trap-assisted thermal generation, trap-assisted tunneling, and band-to-band tunneling [11]. Each of the mechanisms shows distinguishable temperature and field dependences. This requires the band gap energy of the SPAD absorber to be larger than the thermal energy k_bT for a lower rate of thermal carrier generations. Another possible solution is to operate SPADs at lower temperatures, which can significantly reduce the thermal generation of carriers [12].

Afterpulsing (AP) is a secondary phenomenon associated with (DCR) that can lead to the SPAD recording of false detections. In this case, the macroscopic effect is quite different, as a train of pulses could be observed following a single-photon absorption. The physical origin of this phenomenon relies in the trapping of carriers during a previous avalanche event; such trapped carriers can be subsequently released either when the SPAD is reverse biased below or above the breakdown voltage, and in the second case a second spurious avalanche correlated to a former count is registered.

Dead time (DT) is the sum of hold-off and reset times. It represents the minimum time between two consecutive detections. The dead time also sets the maximum achievable count rate of the device, provided that the latter is not limited by other factors like power dissipation. The **hold-off time** is the time that the quenching circuit takes to lower the bias below the breakdown voltage. The reset time is time required to recharge the SPAD to its operating voltage.

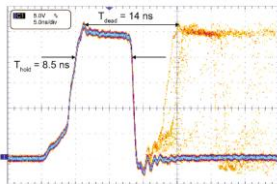


Fig. 3. A voltage waveform example recorded at the anode of the SPAD after the avalanche is sensed by the AQC; T_{hold} is the hold-off time, while T_{dead} is the dead time.

Timing jitter (TJ) is the timing uncertainty between the arrival of the photon at the SPAD and the detection of the avalanche pulse by the electronic circuit [4]. This variability arises from the stochastic nature of carrier generation, transport, and avalanche multiplication processes within the SPAD [12]. The total jitter combination can be calculated from the jitter from the stochastic nature of the avalanche buildup process, the jitter contribution from the diffusion of carriers in the absorption region and the jitter from electronic noise in the readout circuitry.

Photon number resolution (PNR) is important for specific applications such as quantum computation and quantum optics. SPADs may operate in **free-running** mode (always sensitive), or in **time-gated** mode (armed only during certain time windows). However, with conventional techniques of these modes, the SPAD is believed not to have the capability of PNR since avalanches always reach the saturation stage due to relatively long quenching time [9]. Operated in such a fashion, the SPAD can only resolve between zero photons and non-zero photons.

IV. QUENCHING CIRCUITS FOR SPADS

After stage (2) in Figure 1b, the detector is completely blind, meaning that the absorption of another photon does not significantly change the amount of current flowing into the SPAD and therefore it cannot be detected by the electronics. And the avalanche will not stop on its own, external circuit able to quench the avalanche current by lowering the voltage below the breakdown level (3) is necessary. The **quenching element** can be a simple passive (PCR) in series resistor in the order of 100 k Ω or a more complex circuit such as an active quenching circuit (AQC) [2]. In the end, the SPAD must be reset to its initial bias condition (1) to make it ready to detect another photon.

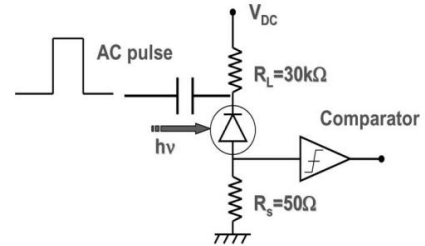


Fig. 4. Schematic of gated quenching circuit [4].

To reduce dead time, AQCs are employed. These circuits can shorten the dead time to a few nanoseconds by dynamically controlling the quenching and reset processes. In AQR circuits, the dead time is determined by the quenching time, programmable hold-off time (which helps reduce afterpulsing by allowing trapped carriers to recombine), and reset time [12].

Passive quenching is the fundamental approach for free-running single-photon detection [9]. However, there are still two technical problems to be solved. One is long recovery time and the associated baseline shift due to avalanche pileups during recovery. The other is large afterpulse probability. The recovery problem could be overcome using a resetting circuit

in practice. Thus, the afterpulsing problem remains the key challenge in implementing free-running detectors. Simply increasing the hold-off time severely limits the count rate.

V. APPLICATIONS

A. Optical Communications

In free-space optical communications, particularly in low-photon-flux regimes, SPADs provide the necessary sensitivity for reliable signal detection. For different absorption coefficients and electromagnetic spectrum ranges, there are several semiconductors that can be used as absorbers for the SPADs.

Silicon (Si) platform is usable for the visible spectral region from 400-1000 nm. Developed Silicon fabrication technique, from the standard CMOS technology makes higher yield production of quality Si SPADs with PDE > 50% and DCR < 1 kHz. Below 400 nm wavelength, absorption is too shallow, while above 1.0–1.1 μm , photons are not absorbed efficiently. Engineering options include shallow and deep junctions, anti-reflection coatings, and back-illumination. Room-temperature operation is standard, but mild thermoelectric cooling is sometimes used to reduce DCR.

InGaAs/InP platform extends SPADs sensitivity to single-photons in the near infrared region of ~900-1700 nm, including telecommunication bands in the 1310 nm and 1550 nm [3]. However, because of the high DCR in InGaAs, and because of after-pulsing caused by the thermal release of trapped avalanche carriers. After the initial avalanche pulse, device areas are typically limited to < 1000 mm^2 and a minimum dead time on the order of 1 ms. Devices are usually Peltier-cooled (~80 °C to -20 °C) and often time-gated [3]. This reduces the DCR, but increases the lifetime of trapped carriers, which results in increased dead time before the detector can be rebiased above breakdown. For example, an InGaAs Geiger-mode avalanche photodiode with a 20 mm diameter might have a 100 kHz false count rate at -50 °C with a 3 ms dead time, resulting in a maximum count rate of less than 300 kHz. Overall, this substantially lowers the PDE (~30-40%).

Ge-on-Si absorbers are still in development and used only for early products as having high margins of errors in fabrication technology, to achieve sensitivity of higher wavelength spectrum ~1300-1600 nm. It is suitable for integrating with silicon photonic integrated circuits. The performance is generally below InGaAs SPADs, however improving over time.

B. Quantum Technologies

SPADs can be used for the detection of entangled photons, in quantum communications with quantum key distribution (QKD), and for fundamental tests of quantum mechanics. Their single-photon sensitivity and fast timing resolution make them well suited for time-correlated single-photon counting in quantum experiments. Particularly, InGaAs/InP SPADs have been widely used for optical fiber quantum communication [9]. InGaAs/InP SPADs operating in low-frequency gating mode were foundational to the development of fiber-based quantum key distribution (QKD). These detectors were first employed in the late 1990s within auto-compensating "plug-and-play" system architectures. Subsequent years saw rapid progress in transmission distance enabled by improvements in SPAD technology. This included the demonstration of plug-and-play

QKD over 40 km of fiber, one-way phase-encoding systems over 48 km, and field tests over a 67 km installed fiber link. By utilizing designs such as a double-SPAD scheme and detectors with low dark count rates, the achievable distance was progressively extended to 100 km and ultimately to 122 km [9].

C. Other Major Implementations

SPADs can be profitably used in a wide range of other applications such as time-resolved spectroscopy; chemistry, physics, and biology; fluid velocimetry; laser ranging; optical time-domain reflectometry; single molecule detection, astronomy; distributed sensing; and optical modulators; investigations of quantum-mechanical phenomena; and studies of high field properties of semiconductors [5].

VI. CONCLUSION

The ability to generate, detect and manipulate quanta of light is the essential ground of quantum applications. In this scenario, a great research effort has been devoted to the investigation of materials and solutions to design single-photon detectors able to provide a combination of high PDE, low DCR and high timing precision. Over the years, SPADs have been studied and continue to develop extensively.

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